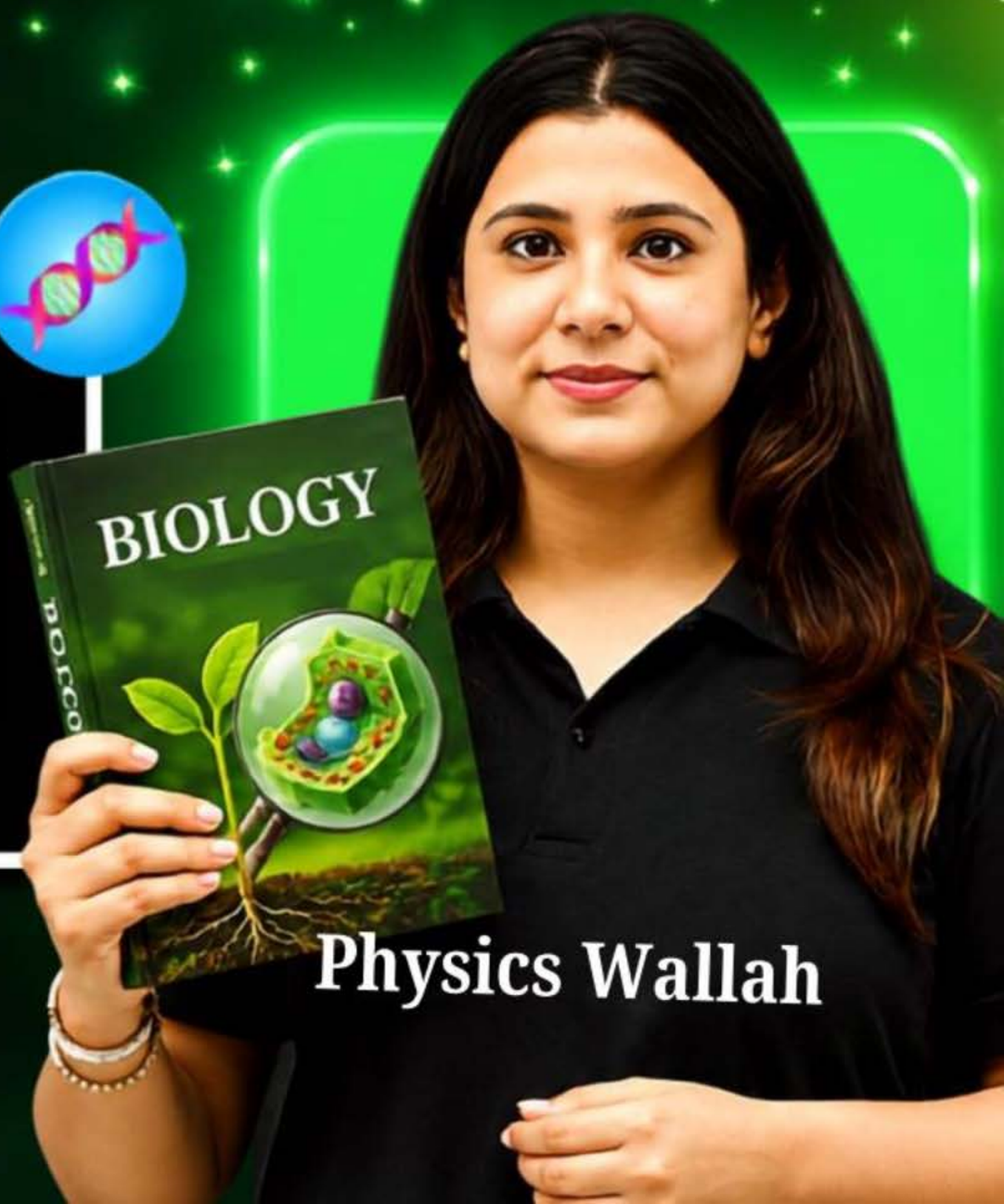
Victory 2027

CLASS 10TH ICSE

THE CIRCULATORY SYSTEM

LECTURE-3 BIOLOGY

By- Upasana Ma'am



Topics To Be Covered

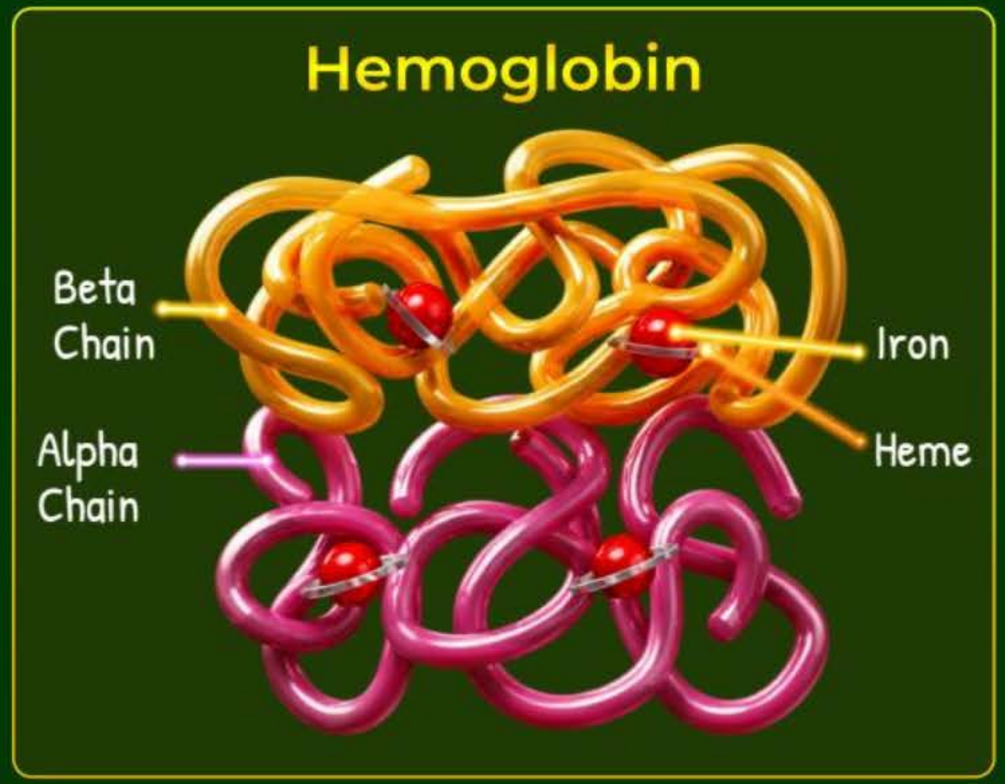
1. RBCs, WBCs.
2. Inflammation


HELLO
NINJAS
忍者参上
影の中で戦う
忍者
忍術最強
忍者
忍者参上

HAEMOGLOBIN

Respiratory System

- **HAEMOGLOBIN** is the effective chemical constituent of RBCs. The red blood cells have a colourless spongy body or **stroma** which contains a **respiratory pigment** haemoglobin (Hb).
- The haemoglobin is formed of an iron containing part **(haemin)** and a protein **(globin).**
- It has the ability to combine readily with oxygen to form **oxyhaemoglobin**, an unstable compound which readily gives up oxygen to the needy tissues.



Q. Hb when combined with CO_2 ____

A) Oxyhaemoglobin

CO.Hb B) Carboxyhaemoglobin

C) Carbaminohaemoglobin ✓

D) NOTA

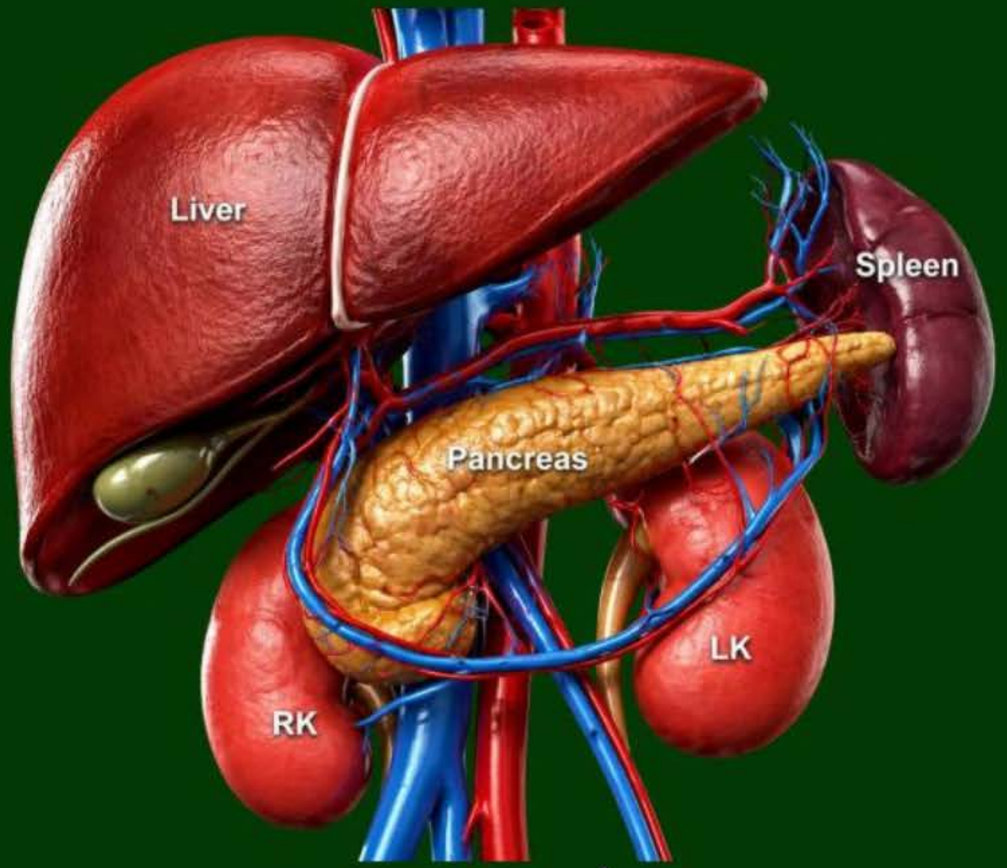
LIFE AND DEATH OF RBCS

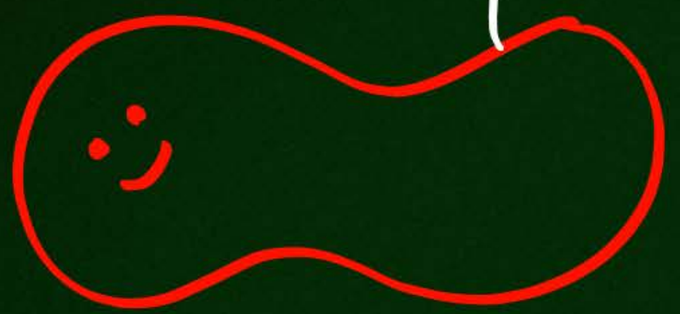
- In adults, marrow of long bones, especially in the ribs, breast bone and ilium of hip girdle.

- In an embryo, in the liver and spleen.
- In children, in bone marrow of all bones until 5 years of age.

- The mature red blood cells are enucleated, but when they are being produced, they have one.

- The average life of an RBC is about 120 days.
- The old and weak red blood cells are destroyed in the spleen, liver and bone marrow; their iron part is retained in the liver while the rest is excreted as a bile pigment (bilirubin).
- Approximately 2 million RBCs are destroyed every second or about 1 per cent of the total erythrocytes in the body are destroyed every-day.

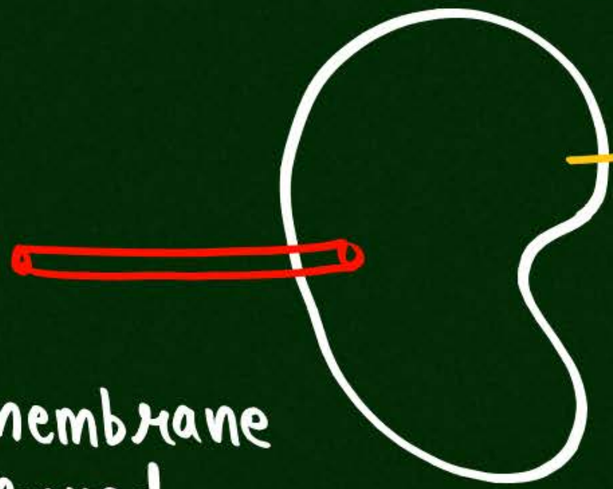








plasma membrane destroyed
→ working efficiency is reduced



Haemo → Iron → liver

Globin → breaks into amino acids
(protein)

More about the number of RBCs :

- New-born infants have a larger number of RBCs about 6-7 million per cubic millimetre (1 mm^3).
- People living at a height of 4,200 m and above, increase their RBCs by nearly 30%.
- Abnormally increased number of RBCs is called Polycythaemia, and their abnormally decreased number is known as Erythropenia.

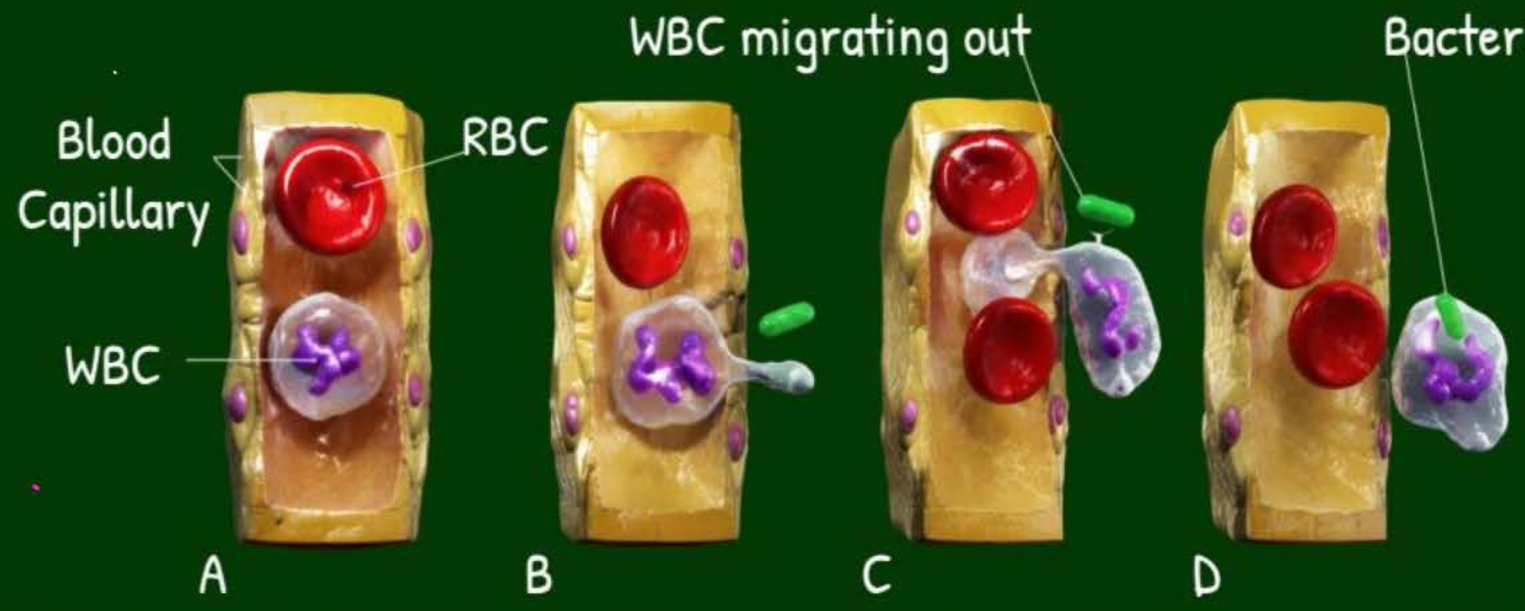


	-cytosis (increased number of cells)	-penia (decreased number of cells)
RBCs (Erythro)	Erythrocytosis Polycythaemia	Erythropenia
WBCs (Leuco)	Leucocytosis Leukemia	Leucopenia
Platelets (Thromo)	Thrombocytosis	Thrombopenia

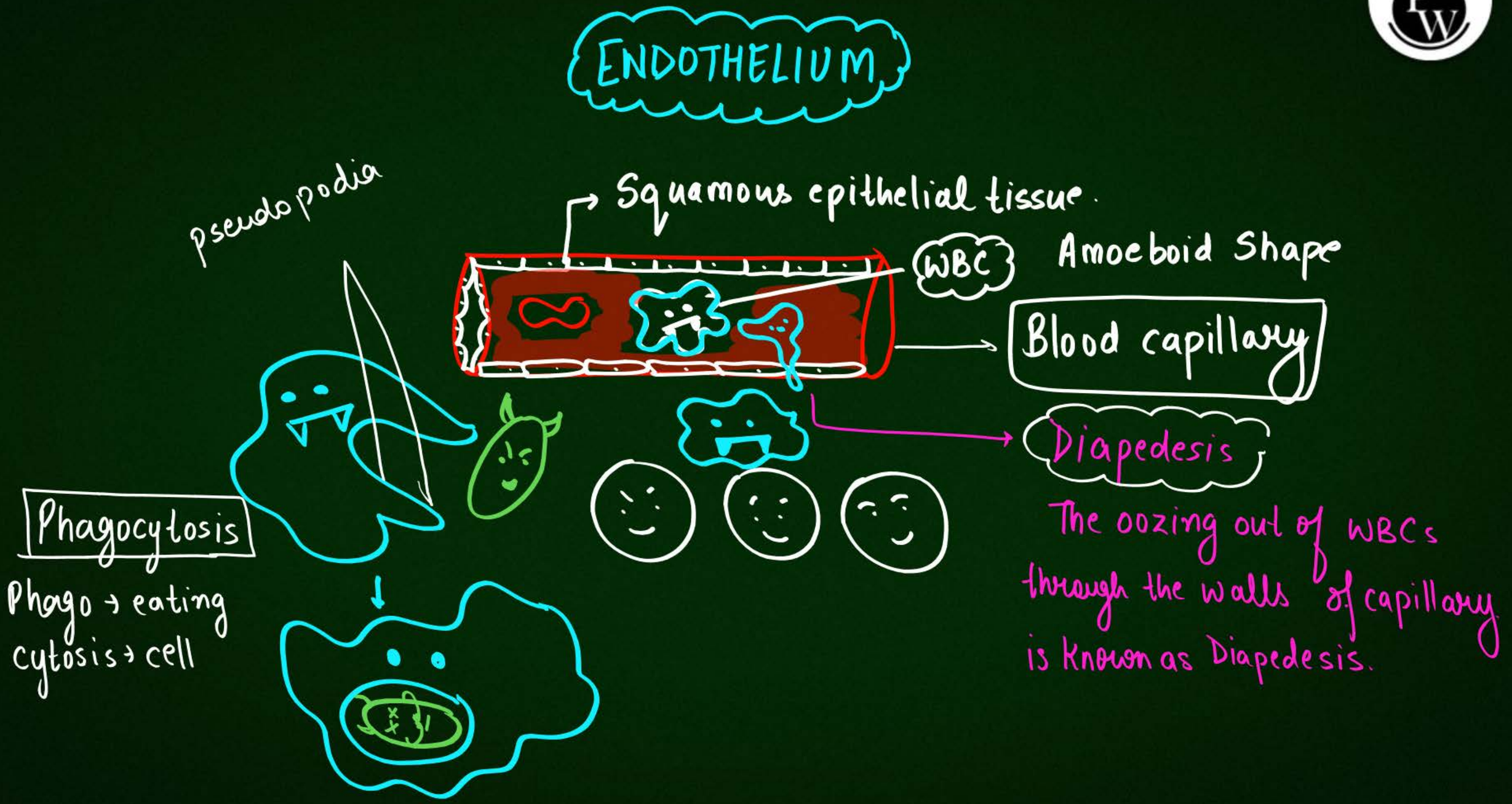
ORIGIN AND LIFE OF WBCS

WBCs (leucocytes)

- Most WBCs are amoeboid and can produce pseudopodia (false feet) with which they can squeeze through the walls of the capillaries into the tissues (diapedesis dia : across, pedesis : oozing out).
- Based on shape and other characteristics, the white blood cells are classified into two major categories (granular and non-granular)



ENDOTHELIUM



2022

Q. The mineral element of Haemoglobin is

A) Ca

B) Mg

C) Fe

D) Na

Cancer ✓

Q. Increased number of RBCs is _

A) Erythropenia

B) Erythrocytosis ✓

C) Polycythaemia.

D) Both B & C.



ORIGIN AND LIFE OF WBCS

- The WBCs are produced in red bone marrow, lymph nodes and sometimes even in liver and the spleen.

Types of WBCs

Granulocytes



granules in cytoplasm.

Agranulocytes



Without granules (Non-granulocytes)

BhE N- G Granulocytes

B → Basophils [Basic dye]

E → Eosinophils (Acid)/Acidophils [Acidic dye]

N → Neutrophils Neutral dye.

MLA

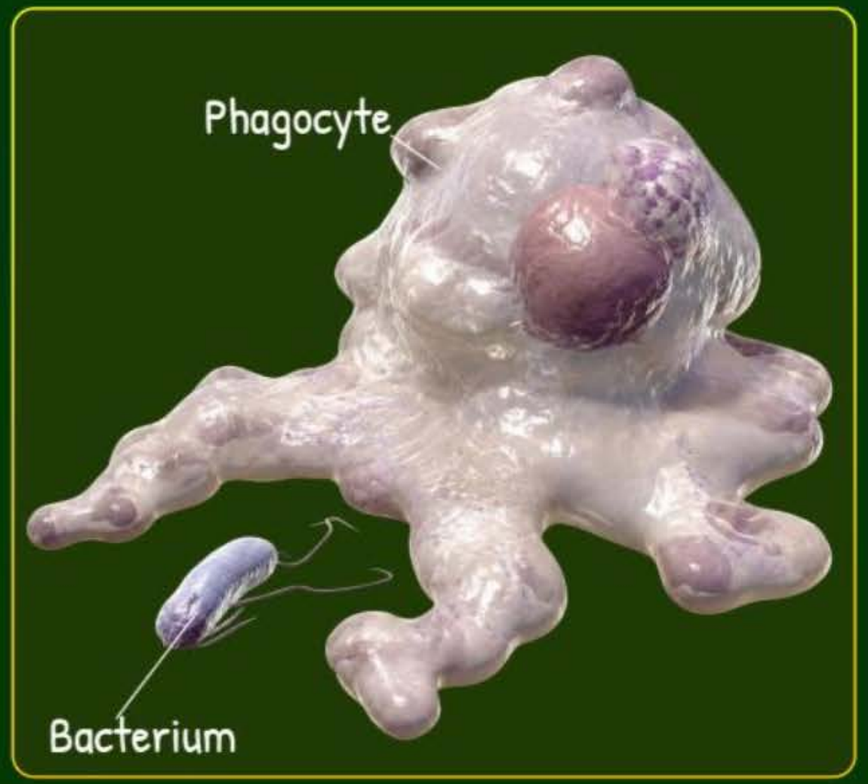
M → MONOCYTES

L → LYMPHOCYTES

FUNCTIONS OF LEUKOCYTES (WBCS) – BODY DEFENCE

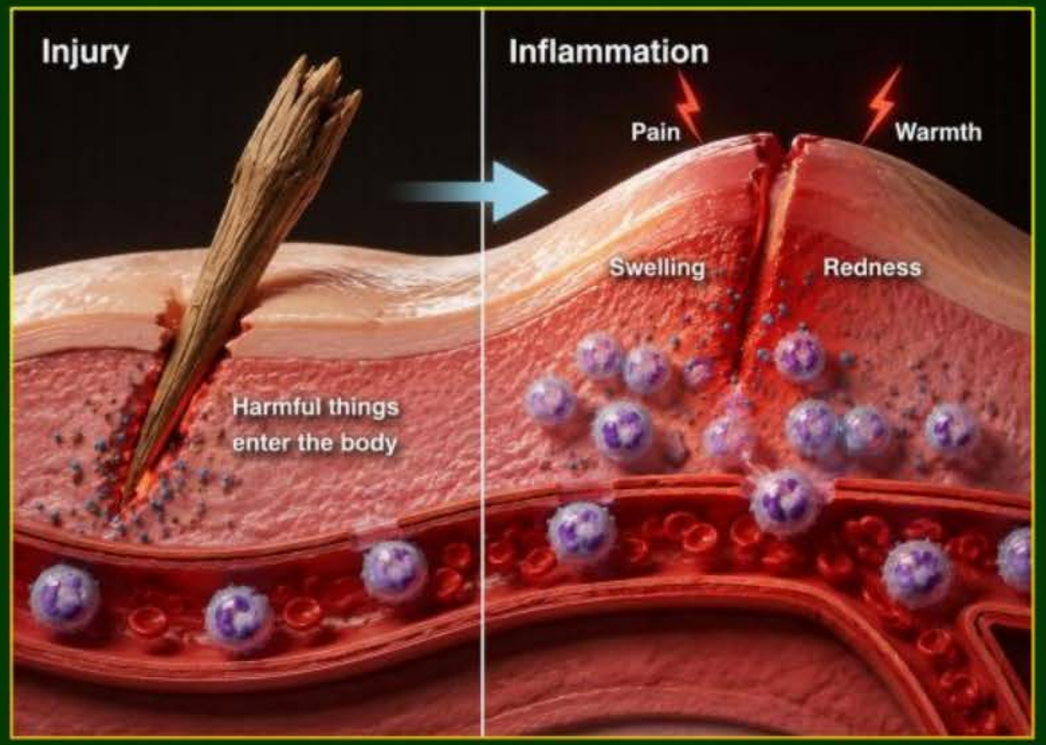
- **Phagocytosis:** This is a process in which most WBCs and particularly the neutrophils engulf particle-like solid substances, especially bacteria.

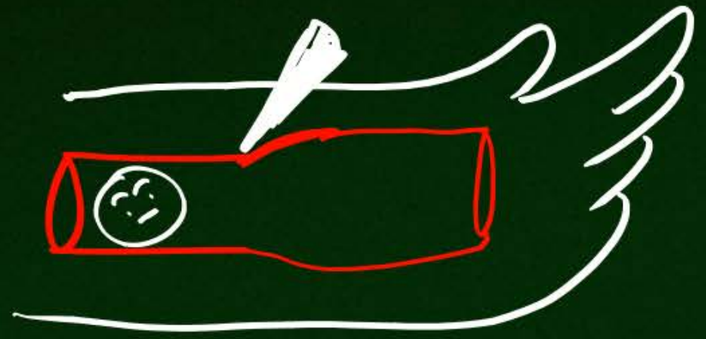
  - An abnormal increase in WBC count up to about 50,000 or more per cubic mm indicates some infection in the body.



INFLAMMATION

- Inflammation occurs due to the reaction of tissues to injury and to localized invasion of germs. The inflamed spot has several characteristics: increased local heat, redness, swelling, pain, etc.







Redness → Increased Blood Supply

Swelling → Dilation of Blood Vessel / Vasodilation

Pus → dead WBCs, bacterias and tissue.

Q. Basophils belong to ____

A) Granulocytes

B) Agranulocytes

BEN-G.

phagocytosis

Q Neutrophils belong to ____

A) Granulocytes

B) Agranulocytes

BEN-G

Thank You